

AI Restaurant Recommendation Agent

Simple Problem Formulation and Environment Classification

1. Problem Formulation

Element	Description
Initial State	The agent receives diner requirements such as favorite cuisine, occasion, budget, dietary restriction, and preferred location. It has no prior assumption about which restaurants are most suitable.
Actions	Collect and organize diner requirements; search restaurant listings; filter restaurants by cuisine, budget, and occasion; compare candidate restaurants; rank suitable restaurants; generate recommendations; and update recommendations using diner feedback.
Transition Model	When the agent analyzes new restaurant listings or receives diner feedback, its knowledge of suitable restaurants, rankings, and recommendations is updated.
Path Cost	The cost may be measured by search time, number of restaurants reviewed, or effort required. The objective is to use the diner's time efficiently.
Goal Test	The diner receives one or more restaurants that closely match the stated cuisine, occasion, budget, dietary restriction, and other requirements.
State Space	All possible combinations of diner requirements, available restaurant listings, filtered restaurants, ranked restaurants, recommendations, and updated recommendations.
Search Space	All possible sequences of collecting requirements, searching listings, filtering restaurants, comparing options, ranking results, generating recommendations, receiving feedback, and updating the plan.
Agents	Multi-agent. Multiple diners, restaurant owners, delivery platforms, and competing recommendation apps can act within the same restaurant discovery environment.

2. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	The agent only knows the information provided by the diner and available restaurant listings. It cannot see unstated preferences or every factor behind the diner's decision.
Determinism	Stochastic / Non-deterministic	Past information cannot guarantee the outcome. A restaurant may be fully booked, its menu or prices may change, or the diner may reject the recommendation.
Episodes	Sequential	Earlier searches and diner feedback affect later recommendations, so decisions are connected rather than isolated.
Change	Dynamic	The restaurant market keeps changing independently. New restaurants open, menus and prices change, and availability shifts.
Values	Discrete & Continuous	Cuisine type and occasion are discrete, while values such as price and distance can be continuous.

Property	Classification	Why
Agents	Multi Agent	Multiple diners, restaurant owners, delivery platforms, and competing recommendation apps can act in the same market at the same time.

3. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The agent moves through many possible states: diner requirements, restaurant listings, filtered restaurants, ranked restaurants, recommendations, and updated recommendations.

Final Answer

The AI Restaurant Recommendation Agent is best classified as a Multi-state agent because it passes through multiple changing states while processing diner requirements, analyzing restaurant listings, filtering and ranking suitable options, generating recommendations, and updating them using diner feedback. It is also partially observable and stochastic because restaurant availability and diner satisfaction cannot be fully predicted in advance.